

CURRICULUM VITAE

NAME

Cabral, Joana

POSITION TITLE

Assistant Professor / Principal Investigator

ORCID

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ELECTRONIC ADDRESS

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EDUCATION/TRAINING

(Begin with baccalaureate or other initial professional education and include postdoctoral training.)

INSTITUTION AND LOCATION	DEGREE	YEAR(s)	FIELD OF STUDY
Universidade Nova de Lisboa, Portugal	BSc/MSc	2002-2007	Biomedical Engineering
Universitat Pompeu Fabra, Barcelona, Spain	PhD	2008-2012	Computational Neuroscience
Universitat Pompeu Fabra, Barcelona, Spain	Postdoc	2012-2014	Computational Neuroscience
University of Oxford, UK	Postdoc	2014-2017	Computational Neuroscience
Universidade do Minho, Braga, Portugal	Assistant Researcher	2017-2025	Computational Neuroscience

A. Personal Statement

Joana Cabral (MSc Biomedical Engineering, PhD Computational Neuroscience) is Assistant Professor in the Department of Bioengineering and the Institute for Systems and Robotics (ISR-Lisboa/LARSyS) at Instituto Superior Técnico (IST), Universidade de Lisboa, where she holds a tenure track position under the FCT Tenure+ Programme. She leads her team at the Evolutionary Systems and Biomedical Engineering Lab (LaSEEB).

Her research focuses on understanding the neural mechanisms underlying large-scale brain dynamics, with emphasis on how oscillatory brain dynamics organise functional brain networks across states of health and disease. She developed LEiDA (Leading Eigenvector Dynamics Analysis), a method now used by over 50 research groups worldwide, and the Metastable Oscillatory Modes (MOMs) framework. She has visiting research affiliations at Champalimaud Foundation and University of Oxford.

Joana Cabral was awarded the L'Oréal UNESCO Award for Women in Science (Portugal) in 2019. She has authored over 70 peer-reviewed publications and supervised students at MSc and PhD level across multiple institutions.

B. Positions and Honors

Positions and Employment

2008-2012 PhD Researcher, Computational Neuroscience Group, Universitat Pompeu Fabra, Barcelona, Spain.

2012-2014 Postdoctoral Researcher, Universitat Pompeu Fabra, Spain (ERC Advanced project).

2014-2017 Postdoctoral Researcher, University of Oxford, Oxford, UK (ERC Consolidator project).

2017-2022 Assistant Researcher, Life and Health Sciences Research Institute (ICVS), Universidade do Minho, Braga, Portugal.

2022-2025 La Caixa Junior Leader Fellow (BRAINSTIM), Life and Health Sciences Research Institute (ICVS), Universidade do Minho, Braga, Portugal.

2017-present Visiting Researcher, University of Oxford, UK.

2018-present Visiting Researcher, Champalimaud Foundation, Lisbon, Portugal.

2025-present Assistant Professor (FCT Tenure+), Department of Bioengineering and ISR-Lisboa/LARSyS, Instituto Superior Técnico, Universidade de Lisboa, Portugal.

2022-present Principal Investigator, LaSEEB, ISR-Lisboa/LARSyS, Lisbon, Portugal.

Honors and Awards

2008-2012 FCT Doctoral Grant, Portugal.

2019 L'Oréal UNESCO Award for Women in Science, Portugal.

Professional Memberships and Editorial Boards

- Reviewer for Nature Neuroscience, Science Advances, PNAS, among others (>70 reviews).
- Member of the Organization for Human Brain Mapping (OHBM).
- Co-PI, BrainStimPredict project (Arqus Research Alliance), with Samir Suweis (Padova) and Miguel Muñoz (Granada).

C. Selected Peer-Reviewed Publications (in chronological order)

- Cabral J, Hugues E, Sporns O, & Deco G (2011). Role of local network oscillations in resting-state functional connectivity. *NeuroImage*, 57(1), 130–139. [640 citations]
- Cabral J, Hugues E, Kringelbach ML, & Deco G (2012). Modeling the outcome of structural disconnection on resting-state functional connectivity. *NeuroImage*, 62(3), 1342–1353. [212 citations]
- Cabral J, Kringelbach ML, & Deco G (2014). Exploring the network dynamics underlying brain activity during rest. *Progress in Neurobiology*, 114, 102–131. [453 citations]
- Cabral J, Luckhoo H, Woolrich M, Joensson M, Mohseni H, Baker A, et al. (2014). Exploring mechanisms of spontaneous functional connectivity in MEG: how delayed network interactions lead to structured amplitude envelopes of band-pass filtered oscillations. *NeuroImage*, 90, 423–435. [371 citations]
- Cabral J, Vidaurre D, Marques P, Magalhães R, Silva Moreira P, et al. (2017). Cognitive performance in healthy older adults relates to spontaneous switching between states of functional connectivity during rest. *Scientific Reports*, 7(1), 5135. [493 citations]
- Cabral J, Kringelbach ML, & Deco G (2017). Functional connectivity dynamically evolves on multiple time-scales over a static structural connectome: Models and mechanisms. *NeuroImage*, 160, 84–96. [457 citations]
- Lord LD, Expert P, Atasoy S, Roseman L, Rapuano K, Lambiotte R, Nutt DJ, Deco G, Carhart-Harris RL, Kringelbach ML, & Cabral J (2019). Dynamical exploration of the repertoire of brain networks at rest is modulated by psilocybin. *NeuroImage*, 199, 127–142. [278 citations]
- Stevner ABA, Vidaurre D, Cabral J, Rapuano K, Nielsen SFV, et al. (2019). Discovery of key whole-brain transitions and dynamics during human wakefulness and non-REM sleep. *Nature Communications*, 10(1), 1035. [299 citations]
- Vohryzek J, Deco G, Cessac B, Kringelbach ML, & Cabral J (2020). Ghost attractors in spontaneous brain activity: recurrent excursions into functionally-relevant BOLD phase-locking states. *Frontiers in Systems Neuroscience*, 14, 20. [150 citations]
- Shine JM, Müller EJ, Munn B, Cabral J, Moran RJ, & Breakspear M (2021). Computational models link cellular mechanisms of neuromodulation to large-scale neural dynamics. *Nature Neuroscience*, 24(6), 765–776. [281 citations]
- Anyaeji CI, Cabral J, & Silbersweig D (2021). On a quantitative approach to clinical neuroscience in psychiatry: lessons from the Kuramoto model. *Harvard Review of Psychiatry*, 29(4), 318–326. [20 citations]
- Cabral J, Castaldo F, Vohryzek J, Litvak V, Bick C, Lambiotte R, Friston K, et al. (2022). Metastable oscillatory modes emerge from synchronization in the brain spacetime connectome. *Communications Physics*, 5(1), 184. [130 citations]
- Luppi AI, Cabral J, Cofre R, Destexhe A, Deco G, & Kringelbach ML (2022). Dynamical models to evaluate structure–function relationships in network neuroscience. *Nature Reviews Neuroscience*, 23(12), 767–768. [44 citations]
- Hancock F, Rosas FE, McCutcheon RA, Cabral J, Dipasquale O, et al. (2023). Metastability as a candidate neuromechanistic biomarker of schizophrenia pathology. *PLOS ONE*, 18(3), e0282707. [32 citations]
- Cabral J, Fernandes HM, & Shemesh N (2023). Intrinsic macroscale oscillatory modes driving long-range functional connectivity in female rat brains detected with ultrafast fMRI. *Nature Communications*, 14, 375. [61 citations]

- Hancock F, Rosas FE, Luppi AI, Zhang M, Mediano PAM, Cabral J, et al. (2025). Metastability demystified: the foundational past, the pragmatic present and the promising future. *Nature Reviews Neuroscience*, 26(2), 82–100. [87 citations]
- Cabral J & Sun H (2026). Beyond Networks: Functional Connectivity as Eigenmode Resonance. *Aperture Neuro*.

Bibliometric indicators (Google Scholar): Total citations 7,454; H-index 38; i10-index 63.

D. Research Support

CSF-MIND – Cerebrospinal fluid dynamics and brain network organisation (2026-2028)

Principal Investigator; 2CA-Braga

BrainStimPredict – Arqus Research Alliance (2025-2027)

Co-Principal Investigator; Arqus Research Alliance

BRAINSTIM – La Caixa Junior Leader Fellowship (2022-2025)

Principal Investigator; La Caixa Foundation

PATHOCONNECTOMICS – CEEC FCT (2019-2022)

Principal Investigator; Portuguese Foundation for Science and Technology